ABHISHEK KUMAR GUPTA

FULL-STACK DEVELOPER & AI SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

Engineering-focused Full-Stack Developer specializing in resilient web platforms, scalable backend architectures, and intelligent multi-agent AI systems. Proven contributor to GDG Ranchi and open-source initiatives, skilled in designing RESTful APIs, real-time WebSocket pipelines, and LLM orchestration (LangChain, LangGraph, Gemini).

TECHNICAL COMPETENCIES

Languages & Runtimes:	TypeScript, JavaScript (ES6+), Python, Node.js, SQL, HTML5, CSS3
Backend & Real-Time:	Express.js, FastAPI, RESTful APIs, WebSockets (Socket.IO), JWT, Webhooks, Microservices
Frontend & UI:	React.js, Next.js, Tailwind CSS, Redux Toolkit, GSAP, Responsive Design, Vite
Databases & Cloud:	PostgreSQL, MongoDB, Redis (Pub/Sub & Caching), Docker, RabbitMQ, Cloudflare Edge CDN
AI & Agentic Systems:	LangChain, LangGraph (Stateful DAGs), Google Gemini 2.5 API, Qdrant Vector DB, RAG Pipelines
Tools & Workflows:	Git, GitHub, Postman, Linux Environment, npm, Vite, CI/CD Fundamentals

EXPERIENCE & COMMUNITY LEADERSHIP

Core Tech Team Member | Google Developer Group (GDG) Ranchi — 2024 - Present

- Engineered software platforms for community tech events, including the official DevFest Ranchi portal handling 5,000+ requests.
- Implemented message queue buffering with RabbitMQ to handle ticket checkout concurrency spikes with zero service drops.
- Led technical developer workshops and hands-on activities focusing on Gemini CLI, API workflows, and modern web frameworks.
- Collaborated across cross-functional organizing teams to coordinate hackathons, speaker platforms, and attendee verification.

Open Source Contributor | GirlScript Summer of Code (GSSoC) — 2024

- Collaborated on active production open-source repositories; identified edge-case bugs, proposed solutions, and submitted clean PRs.
- Strengthened practical knowledge of Git branching workflows, test coverage verification, and peer code review protocols.

FEATURED ENGINEERING PROJECTS

CIITM Student Management System | MERN, Socket.IO, Redis, Razorpay Gateway

Role-based academic management platform serving students, faculty, and administrators.

- Engineered REST APIs with Express.js and modular MVC structure; enforced strict Role-Based Access Control (RBAC).
- Integrated Razorpay Payment Gateway with cryptographically verified HMAC SHA256 webhooks for atomic fee reconciliation.
- Built room-partitioned bidirectional real-time notifications via Socket.IO, reducing broadcast overhead by 70%.

NEXORA Autonomous Multi-Agent AI | React, Python, FastAPI, LangGraph, Qdrant

Stateful multi-agent swarm platform orchestrating autonomous decomposition of long-horizon software engineering tasks.

- Architected core agent supervisor engine utilizing LangGraph directed acyclic graph (DAG) state machines.
- Integrated Qdrant vector database for semantic memory retrieval across long sessions; enforced Pydantic schemas for safe execution.
- Constructed real-time WebSocket DAG visualizer in React tracking thought loops, verification steps, and artifacts.

Tolkify Real-Time Messaging Platform | React, TypeScript, Node.js, Socket.IO, Redis

Distributed real-time communication platform engineered for sub-second delivery latency and zero perceptual delay.

- Designed typed WebSocket event handlers across server and client with optimistic state updates in Zustand.
- Integrated @socket.io/redis-adapter allowing horizontal clustering across multiple Node.js instances with zero message loss.

AI Movie Recommender | React, Vite, Google Gemini 2.5 Flash, TMDB API

Intelligent movie discovery engine utilizing Gemini 2.5 structured JSON output mode to map subjective prompts to cinema catalog.

- Crafted few-shot prompt contracts and parallelized TMDB enrichment with Promise.allSettled, cutting latency by 60%.

EDUCATION

Bachelor of Computer Applications (BCA) — 2024 - 2027 (Expected)

Chhotanagpur Institute of Information Technology & Management (CIITM) Dhanbad | MCNU Bhopal

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Web Engineering, Software Architecture.